

Maxine Elena Calle

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EMPLOYMENT AND EDUCATION

Brown University Tamarkin Assistant Professor <i>NSF Postdoctoral Research Fellow</i> – Postdoc mentor: Thomas Goodwillie	Providence, RI 2027–2031
University of Bonn Hausdorff Center for Mathematics postdoctoral scholar – Postdoc mentor: Stefan Schwede	Bonn (Germany) 2026–2027
University of Pennsylvania Ph.D. in Mathematics <i>NSF Graduate Research Fellow</i> – Thesis advisor: Mona Merling – Thesis title: “Algebraic K -theory of orbispaces” M.A. in Mathematics – Thesis advisor: Jonathan Block – Thesis title: “An Introduction to Symplectic Geometry for Lagrangian Floer Homology”	Philadelphia, PA 2026
Reed College B.A. in Mathematics, concentration in Philosophy <i>Phi Beta Kappa</i> – Thesis advisor: Kyle Ormsby – Thesis title: “Morse Theory and Flow Categories”	Portland, OR 2020

RESEARCH INTERESTS

Algebraic topology, homotopy theory, and category theory, with a focus on higher algebraic K -theory, equivariant and stable homotopy theory, and applications to manifolds.

PAPERS AND PUBLICATIONS

Research publications

- *A comparison of definitions of equivariant trees*. Joint with J.E. Bergner, D. Chan, A. Osorno, and M. Sarazola. 2025. 31 pages. Available at [arXiv:2512.14956](https://arxiv.org/abs/2512.14956). *To appear in the Proceedings of the 2024 Mathematics Research Community on Homotopical Combinatorics*.
- *Segal K -theory factors through Waldhausen categories*. Joint with D. Chan. *Proceedings of the American Mathematical Society*: Vol. 154, No. 10 (2026), pp.4149–4166.
- *Squares K -theory and 2-Segal spaces*. Joint with M. Sarazola. *Annals of K-Theory*: Vol. 11, No. 2 (2026), pp.261–308.
- *A genuine G -spectrum for the cut-and-paste K -theory of G -manifolds*. Joint with D. Chan. 2026. 15 pages. *Bulletin of the London Mathematical Society*, Vol. 58, No. 4: e70352. 16 pages.
- *Equivariant Trees and Partition Complexes*. Joint with J.E. Bergner, P. Bonventre, D. Chan, and M. Sarazola. 2026. *Theory and Applications of Categories*: Vol. 45, 2026, No. 15, pp.501–536.
- *The spectrum of the Burnside Tambara functor*. Joint with D. Chan, D. Mehrle, J.D. Quigley, B. Spitz, and D. Van Niel. 2026. *International Mathematics Research Notices*: Vol. 2026, Iss. 2, rna388. 26 pages.

- *A linearization map for genuine equivariant algebraic K-theory*. Joint with D. Chan and A. Mejia. 2023. 50 pages. Available at arXiv:2309.08025. *To appear in Algebraic & Geometric Topology*.
- *Nested cobordisms, Cyl-objects and Temperley-Lieb algebras*. Joint with R.S. Hoekzema, L. Murray, N. Pacheco-Tallaj, C. Rovi, and S. Sridhar-Shapiro. 2025. *Topology and its Applications*: Vol. 376, 109448. 36 pages.
- *A combinatorial K-theory perspective on the edge reconstruction conjecture in graph theory*. Joint with J. J. Gould. 2025. *Homology, Homotopy, and Applications*: Vol. 27(1), pp.293–322.
- *The Spectrum of the Burnside Tambara Functor of a Cyclic Group*. Joint with S. Ginnett. 2023. *Journal of Pure and Applied Algebra*: Vol. 227, Iss. 8, 107344. 23 pages.
- *The Tambara Structure of the Trace Ideal on Cyclic Groups*. Joint with S. Ginnett, with an appendix by H. Chen and X. Chen. 2020. *Journal of Algebra*: Vol. 560, pp.114–143.
- *Sharp Sectional Curvature Bounds and a New Proof of the Spectral Theorem*. Joint with C. Dunn. 2020. *Involve, a Journal of Mathematics*: Vol. 13, No. 3, pp.445–454.
- *k-Plane Constant Curvature Conditions*. 2019. *Rose-Hulman Undergraduate Mathematics Journal*: Vol. 20, Iss. 2, Article 6. 20 pages.

Preprints

- *On the classifying space of a Morse flow category*. Joint with F. Liu. 2026. 32 pages. Available at arXiv:2603.23695. *Submitted*.
- *Equivariant algebraic K-theory for symmetric monoidal Mackey functors*. Joint with D. Chan and M. Péroux. 2023. 32 pages. Available at arXiv:2312.04705. *Submitted*.

Other publications

- *Looking into Mirror Symmetry at the 2024 JMM*. In *Feature Stories of the American Mathematical Society*, February 2024.
- *Bringing Math into The Conversation: My Summer as a Mass Media Fellow*. In *Notices of the American Mathematical Society*, February 2024, pp.274–275.
- *A brief illustrated guide to ‘scissors congruence’ – an ancient geometric idea that’s still fueling cutting-edge mathematical research*. Co-authored with M. Merling. 2023. *The Conversation U.S.*
- *Proving Fermat’s last theorem: 2 mathematicians explain how building bridges within the discipline helped solve a centuries-old mystery*. Co-authored with D. Bressoud. 2023. *The Conversation U.S.*

FELLOWSHIPS, GRANTS, AND AWARDS

Academic and research

- Oberwolfach Research Fellows program on “Derived orbifold bordism”
Oberwolfach Research Institute for Mathematics 2027
- Carlitz–Zippin Prize for an exceptional Ph.D. thesis
University of Pennsylvania, Mathematics Department 2026
- Mathematical Sciences Postdoctoral Research Fellowship
National Science Foundation
Award deferred (1 year) for a postdoctoral appointment at the University of Bonn 2026
- AIM SQuaREs grant for “Algebras over genuine equivariant operads”
American Institute of Mathematics 2025
- AIM SQuaREs grant for “An algebraic approach to Thom spectra”
American Institute of Mathematics 2025
- AIM SQuaREs grant for “Two-dimensional nested cobordisms and TQFTs with substructure”
American Institute of Mathematics 2024
- Herbert S. Wilf Memorial Award for excellence in research and communication
University of Pennsylvania, Mathematics Department 2023–2024
- Dean’s Scholar
University of Pennsylvania, School of Arts & Sciences 2023
- Graduate Research Fellowship
National Science Foundation 2020–2025

- Calabi Scholar 2020–2025
University of Pennsylvania, Mathematics Department
- Benjamin Franklin & Fontaine Fellowship 2020–2025
University of Pennsylvania
- Phi Beta Kappa 2020
Reed College
- Commendation for Academic Achievement 2016–2020
Reed College

Teaching and pedagogy

- Moez Alimohamed Graduate Student Teaching Award for excellence in teaching 2026
University of Pennsylvania, Mathematics Department
- Prison Teaching Initiative Graduate Fellow 2024–2025
Prison Teaching Initiative at Princeton University
- CETLI Teaching Certificate 2024
Center for Teaching and Learning at the University of Pennsylvania
- Master TA 2023–2026
University of Pennsylvania, Mathematics Department
- Inclusive & Equitable Teaching Fellow 2023–2024
Center for Teaching and Learning at the University of Pennsylvania
- Good Teaching Award, Math 3700 Fall 2022
University of Pennsylvania, Mathematics Department

Service, communication, and outreach

- Postdoctoral/Advanced Graduate Ambassadorship 2023–2024
Lisa Simonyi and the Women and Mathematics Program at the Institute for Advanced Study
- Mass Media Fellow 2023
*American Association for the Advancement of Science
Sponsored by the American Mathematical Society and hosted by The Conversation U.S.*
- Penn Graduate Community-Engagement Research Fellow Summers 2021 and 2022
The Netter Center at the University of Pennsylvania
- Elevating Mathematics competition, honorable mention (joint with L. Williams) 2020
National Academies of Sciences, Engineering, and Medicine
- MAA MathFest Outstanding Presentation Award 2019
Joint Mathematics Meetings

TEACHING EXPERIENCE

The University of Pennsylvania	Philadelphia, PA
Instructor <i>Math 3120: Linear algebra</i>	Summer 2025
Teaching Assistant <i>Math 3700: Algebra</i>	Spring 2026
<i>Math 2600: Honors Calculus II</i>	Spring 2023
<i>Math 3700: Algebra</i>	Fall 2022
Grader <i>Math 6180: Algebraic Topology I</i>	Fall 2025
Workshop Facilitator <i>Trauma-Informed Pedagogy</i>	Spring 2025
<i>Inclusive & Equitable Teaching in Mathematics (mini-course)</i>	Fall 2023
<i>Mathematics Department TA Training</i>	Falls 2023–2025
<i>School of Arts & Sciences Graduate TA Training: Inclusive & Equitable Teaching at Penn</i>	Fall 2023
Reed College	Portland, OR
Individual Tutor and Drop-in Tutor at the Math Help Center <i>Math 111: Calculus, Math 112: Introduction to Analysis, Math 113: Discrete Structures,</i> <i>Math 201: Linear Algebra, Math 202: Vector Calculus, Math 311: Complex Analysis,</i> <i>Math 321: Real Analysis, and Math 332: Abstract Algebra</i>	Spring 2017–Spring 2020

Prison teaching

Co-instructor at State Correctional Institution Chester
Mathematics in the Courtroom

Chester, PA
 Spring 2026

Co-instructor at South Woods State Prison
Math-015: Beginning Mathematics
Math-015: Beginning Mathematics

Bridgeton, NJ
 Fall 2024
 Spring 2022

Co-instructor at Garden State Youth Correctional Facility
Math-015: Beginning Mathematics

Crosswicks, NJ
 Fall 2021

Mathematics Tutor at South Woods State Prison
Math-010: Number Systems, Math-015: Beginning Mathematics,
Math-020: Elementary Algebra

Bridgeton, NJ
 2023–2025

Course Development team leader
Math-015: Beginning Mathematics

Summer 2024

OTHER EXPERIENCE AND SERVICE

Seminar and conference organization

- **Talbot Workshop** 2023–2026
 - Talbot 2026: Topological holography
co-organizer with A. Karapetyan, C. Li, and S. Xue
 - Talbot 2025: Homological stability,
co-organizer with A. Karapetyan and E. Sukarto
 - Talbot 2024: Topological cyclic homology of ring spectra,
co-organizer with A. Holeman, A. Karapetyan, L. Keenan, and E. Sukarto
 - Talbot 2023: Computations in stable motivic homotopy theory,
co-organizer with A. Holeman, L. Keenan, E. Sukarto, and L. Yang
- **Atlantic Meeting on Topology, Representation theory, and K -theory (AMTRaK),** 2024–2025
co-organizer with A. Chedalavada and B. Spitz
 - Homological stability, at the University of Pennsylvania March–April 2025
 - Hermitian K -theory, at the University of Virginia February 2025
 - Dualizable categories and continuous K -theory, at Johns Hopkins University November 2024
 - Equivariant and parametrized homotopy theory, at the University of Pennsylvania September 2024
- **Gender Minorities in Topology and Related Areas Konfernce (GeMTRAK)** April 2024
 at the University of Pennsylvania, *co-organizer with M. Ong and H. Stufflebeam*
- **Graduate Geometry/Topology Seminar** at the University of Pennsylvania, 2022–2023
co-organizer with A. Mejia
- **Graduate Pizza Seminar** at the University of Pennsylvania, *co-organizer with K. Canzius* 2021–2022
- **UP GRADe** workshop at the University of Pennsylvania, *co-organizer with* October 2021
A. Gibney, J. Hartmann, M. Merling, M. Ong, and E. Urheim

Community building and mentorship

- **Directed Reading Program** at the University of Pennsylvania, *co-organizer with*
L. Ferreira Guilhoto (Fall 2023–Spring 2025) and mentor for the following projects:
 - Shotaro Hiranuma, *Continuing in category theory* Spring 2025
Category theory and logic Fall 2024
 - Talia Becker Calazans, *Combinatorial topology* Fall 2023
 - Elena Isasi Theus, *Algebraic topology and the mathematical basis of topological data analysis* Spring 2023
Fundamentals of computational applied topology Fall 2022
 - Cianán Conefrey-Shinozaki, *2D topological quantum field theories* Spring 2022

– Michael Zeng, <i>Simplicial homotopy theory</i>	Fall 2021
• Prison Education Program at Eastern University, <i>volunteer</i>	Spring 2024
• Prison Teaching Initiative at Princeton University, <i>volunteer</i>	2021–2025
• GeMs in Math group at the University of Pennsylvania, <i>co-coordinator</i>	2021–2025
• STEMGeMs at Reed College, <i>co-leader and department liaison</i>	2019–2020
• Peer Mentor Program at Reed College, <i>mentor</i>	2017–2018

Science communication, outreach, and other service

• Penn Grad Talks at Penn Arts & Sciences <i>Natural Sciences finalist</i>	Spring 2026
• Math Circles at West Philadelphia High School <i>Volunteer</i>	Spring 2024
• The Conversation U.S. <i>Freelancer (Fall 2023) and AMS-AAAS 2023 Mass Media Fellow (Summer 2023)</i>	2023
• Math Circles in West Philadelphia project at the University of Pennsylvania <i>Graduate fellow, joint project with M. Ong, mentored by M. Merling and Y. Ou</i>	Summer 2022 Philadelphia, PA
• Mathematics Enrichment and Academically-Based Community Service project at the University of Pennsylvania <i>Graduate fellow mentored by M. Merling</i>	Summer 2021 Philadelphia, PA
• Referee and reviewer <i>Advances in Mathematics, Proceedings of the AMS, PUMP Journal of Undergraduate Research, MathSciNet, zbMATH Open</i>	

SELECTED RESEARCH WORKSHOPS

• AIM Workshop at American Institute of Mathematics <i>Invited participant in “Scissors congruences, algebraic K-theory and Steinberg modules”</i>	July 2024 Caltech, CA
• Women in Topology IV at the Hausdorff Institute <i>Team member for research project on cobordism categories</i>	August 2023 Bonn (Germany)
• RTG Collaborative Workshop in Homotopy Theory <i>Team member for research project on equivariant homotopy theory</i>	August 2022 Lexington, VA
• Collaborative Mathematics Research Group at Reed College <i>Participant in undergraduate research program on algebraic topology</i>	Summer 2019 Portland, OR
• NSF REU program at California State University, San Bernardino <i>Participant in undergraduate research program on differential geometry</i>	Summer 2018 San Bernardino, CA

SELECTED TALKS AND PRESENTATIONS

Research conference talks

Conference of the French Algebraic Topology Network <i>Cut-and-paste K-theory of manifolds and SK-automorphisms</i>	Grenoble, France October 2026
AMS 2025 Fall Eastern Sectional Meeting, Special Session: Homotopy theory <i>Segal K-theory factors through Waldhausen categories</i>	St. Louis, MO October 2025
CMO Workshop: Equivariant Topological Quantum Field Theory <i>Cut-and-paste K-theory of manifolds and cobordism</i>	Oaxaca, Mexico September 2025
Mathematical Congress of the Americas, Special Session: Interactions of equivariant bordism and low dimensional topology <i>Towards algebraic K-theory of orbifolds</i>	Miami, FL July 2025
Joint Mathematics Meetings, Special Session: Homotopical combinatorics <i>The Edge Reconstruction Conjecture and Combinatorial K-theory</i>	Seattle, WA January 2025
Binghamton University Graduate Combinatorics, Algebra, and Topology Conference (BUGCAT) <i>The Edge Reconstruction Conjecture and Combinatorial K-theory</i>	Binghamton, NY October 2024
AMS 2024 Fall Eastern Sectional Meeting, Special Session: Homotopy theory and K-theory <i>Squares K-theory and 2-Segal spaces</i>	Albany, NY October 2024

Graduate Student Conference in Algebra, Geometry, and Topology <i>Equivariant Trees and Partition Complexes</i>	Philadelphia, PA June 2024
Joint Mathematics Meetings, Special Session: Equivariant techniques in stable homotopy theory <i>A linearization map for equivariant A-theory</i>	San Francisco, CA January 2024
Joint Mathematics Meetings, Special Session: Topics in equivariant algebra <i>Ideals in the Burnside Tambara functor on a cyclic group</i>	San Francisco, CA January 2024
Binghamton University Graduate Combinatorics, Algebra, and Topology Conference (BUGCAT) <i>Equivariant Partition Complexes and Trees</i>	Binghamton, NY November 2022
Nebraska Conference for Undergraduate Women in Mathematics <i>The Tambara Structure of the Trace Ideal</i>	Lincoln, NE January 2020
Undergraduate Mathematics Symposium <i>Putting the 'k' in Curvature: k-Plane Constant Curvature Conditions</i>	Chicago, IL November 2018
Northwest Undergraduate Mathematics Symposium <i>Putting the 'k' in Curvature: k-Plane Constant Curvature Conditions</i>	Salem, OR November 2018

Invited seminar talks

eCHT Research Seminar <i>Prime ideals in the Burnside Tambara functor</i>	Online September 2026
MPIM Abstract Homotopy Theory Seminar <i>Group-completion K-theory via Waldhausen categories</i>	Bonn, Germany September 2026
Texas State Topology Seminar <i>Classifying spaces and Morse flow categories</i>	San Marcos, TX May 2026
University of New Mexico Geometry & Topology Seminar <i>Towards algebraic K-theory of orbifolds</i>	Las Cruces, NM March 2026
University of Pennsylvania Geometry & Topology Seminar <i>Nested cobordisms and TQFTs</i>	Philadelphia, PA February 2026
University of Copenhagen Algebra and Topology Seminar <i>Towards algebraic K-theory of orbifolds</i>	Online February 2026
Topological Quantum Field Theory Club Seminar <i>Nested cobordisms and TQFTs</i>	Online January 2026
University of Indiana Bloomington Topology Seminar <i>Cut-and-paste K-theory of manifolds and SK-automorphisms</i>	Bloomington, IN December 2025
Universität Bonn Topology Seminar <i>Towards algebraic K-theory of orbifolds</i>	Bonn, Germany November 2025
Brown University Topology Seminar <i>Cut-and-paste K-theory of manifolds and SK-automorphisms</i>	Providence, RI November 2025
PIMS Collaborative Research Group "Diagram Categories in Homotopy Theory" Seminar <i>Cut-and-paste K-theory of manifolds and SK-automorphisms</i>	Online October 2025
University of Pennsylvania Geometry & Topology Seminar <i>Towards algebraic K-theory of orbifolds</i>	Philadelphia, PA February 2025
University of Chicago Topology Seminar <i>Towards A-theory of orbifolds</i>	Chicago, IL January 2025
Northwestern University Topology Seminar <i>Towards A-theory of orbifolds</i>	Evanston, IL January 2025
University of Virginia Topology Seminar <i>Towards A-theory of orbifolds</i>	Charlottesville, VA November 2024
Johns Hopkins University Topology Seminar <i>Towards A-theory of orbifolds</i>	Baltimore, MD October 2024
Texas State Topology Seminar <i>A linearization map for equivariant A-theory</i>	San Marcos, TX March 2024
University of Indiana Bloomington Topology Seminar <i>Nested Cobordisms and TQFTs</i>	Bloomington, IN March 2024
University of Minnesota Topology Seminar	January 2024

<i>All connective G-spectra come from K-theory</i>	Minnesota, MN
Loyola University TACO Seminar	Chicago, IL
<i>Equivariant Partition Complexes and Trees</i>	November 2023
Michigan State University RTG Seminar	East Lansing, MI
<i>Nested Cobordisms and TQFTs</i>	November 2023
SUNY Binghamton Geometry and Topology Seminar	Binghamton, NY
<i>Nested Cobordisms and TQFTs</i>	September 2023
UCLA Algebraic Topology Seminar	Los Angeles, CA
<i>Equivariant Partition Complexes and Trees</i>	May 2023
UCSD Topology Seminar	San Diego, CA
<i>Equivariant Partition Complexes and Trees</i>	May 2023
Purdue University Topology Seminar	Lafayette, IN
<i>Equivariant Partition Complexes and Trees</i>	April 2023
University of Rochester Topology Seminar	Rochester, NY
<i>Equivariant Partition Complexes and Trees</i>	April 2023
University of Virginia Topology Seminar	Charlottesville, VA
<i>Equivariant Partition Complexes and Trees</i>	February 2023

Invited expository talks

Penn Arts & Sciences Penn Grad Talks	Philadelphia, PA
<i>What is math good for, anyways?</i> (short talk)	March 2026
Moduli spaces of pseudo-holomorphic curves workshop	Otego, NY
<i>Orbispaces as stacks: geometry and examples</i>	May 2025
SLMath Hot Topics Workshop: Life after the Telescope Conjecture	Berkeley, CA
<i>Cyclotomic spectra</i>	December 2024
Talbot 2024: Topological cyclic homology of ring spectra	Nacogdoches, TX
<i>THH as a cyclotomic spectrum</i>	August 2024
Scissors congruences, algebraic K -theory and Steinberg modules AIM Workshop	Pasadena, CA
<i>The scissors congruence K-theory of polytopes is a Thom spectrum</i>	July 2024
Spaces of Manifolds: Algebraic and Geometric Approaches BIRS seminar	Online
<i>The stable parametrized h-cobordism theorem: the K-theory part</i>	October 2023
Scissors Congruence, Algebraic K -Theory, and Trace Methods summer school	Bloomington, IN
<i>Cut-and-paste K-theory of manifolds and square K-theory</i>	June 2023
Talbot 2022: Scissors congruence and algebraic K -theory	New Lebanon, NY
<i>Square K-theory and Manifold Invariants</i>	June 2022

Talks for undergraduates

UPGrade Workshop	Philadelphia, PA
<i>How to cut up shapes using algebraic topology</i> (short talk)	May 2026
Reed College Mathematics Colloquium	Portland, OR
<i>The Edge Reconstruction Conjecture and Combinatorial K-theory</i>	October 2024
Providence College Mathematics Colloquium	Providence, RI
<i>The Edge Reconstruction Conjecture and Combinatorial K-theory</i>	September 2024
Temple University Math Club	Philadelphia, PA
<i>Scissors Congruence: how to cut up shapes and get away with it</i>	March 2024
Penn Undergraduate Mathematics Society (PUMS) speaker series	Philadelphia, PA
<i>Social choice problems and algebraic topology</i> (short talk)	April 2024
<i>Configurations in the plane</i> (short talk)	November 2023
<i>Scissors Congruence: how to cut up shapes and get away with it</i>	December 2022
Penn Graduate Student Open House lightning talks	Philadelphia, PA
<i>A quick introduction to cut-and-paste manifolds</i>	February 2026
<i>A quick introduction to cut-and-paste manifolds</i>	February 2025
<i>Social choice problems and algebraic topology</i>	February 2024